



TALLER DE DESPLIEGUE FULL-STACK



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04

TALLER DE DESPLIEGUE FULL-STACK



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3. Probar conexión con NodeJs





Base de Datos

Ajustes de instalación



1. Eliminar la instalación de Mysql - Mariadb

Ejecutar Terminar:

- Detener los Servicios
 - `sudo systemctl stop mariadb`

```
developer@10: ~  
developer@10:~$ sudo systemctl stop mariadb  
[sudo] password for developer:  
developer@10:~$
```





Base de Datos

Ajustes de instalación



- Deshabilitar los Servicios
 - `sudo systemctl disable mariadb`

```
developer@10: ~  
developer@10:~$ sudo systemctl stop mariadb  
[sudo] password for developer:  
developer@10:~$ sudo systemctl disable mariadb  
Synchronizing state of mariadb.service with SysV service script with /usr/lib/sy  
stemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install disable mariadb  
Removed '/etc/systemd/system/multi-user.target.wants/mariadb.service'.  
developer@10:~$
```



Base de Datos

Ajustes de instalación



- Desinstalar los paquetes de MariaDB
 - `sudo apt remove --purge mariadb-server mariadb-client`

```
developer@10: ~  
developer@10:~$ sudo apt remove --purge mariadb-server mariadb-client  
The following packages were automatically installed and are no longer required:  
galera-4 liburing2  
libcgi-fast-perl mariadb-client-core  
libcgi-pm-perl mariadb-common  
libconfig-inifiles-perl mariadb-plugin-provider-bzip2  
libdbd-mariadb-perl mariadb-plugin-provider-lz4  
libdbi-perl mariadb-plugin-provider-lzma  
libfcgi-bin mariadb-plugin-provider-lzo  
libfcgi-perl mariadb-plugin-provider-snappy  
libfcgi0t64 mariadb-server-core  
libhtml-template-perl mysql-common  
libmariadb3 pv  
libpcre2-posix3 rsync  
libterm-readkey-perl socat  
Use 'sudo apt autoremove' to remove them.  
  
REMOVING:  
mariadb-client* mariadb-server*  
  
Summary:  
Upgrading: 0, Installing: 0, Removing: 2, Not Upgrading: 0  
Freed space: 123 MB  
  
Continue? [Y/n] Y  
(Reading database ... 200256 files and directories currently installed.)  
Removing mariadb-server (1:11.8.3-0+deb13u1) ...  
Removing mariadb-client (1:11.8.3-0+deb13u1) ...  
Processing triggers for man-db (2.13.1-1) ...  
(Reading database ... 200092 files and directories currently installed.)  
Purging configuration files for mariadb-client (1:11.8.3-0+deb13u1) ...  
Purging configuration files for mariadb-server (1:11.8.3-0+deb13u1) ...  
developer@10:~$
```





Base de Datos

Ajustes de instalación



- Desinstalar los paquetes de MariaDB
 - sudo apt autoremove

```
developer@10: ~  
developer@10:~$ sudo apt autoremove  
REMOVING:  
galera-4  
libcgi-fast-perl  
libcgi-pm-perl  
libconfig-inifiles-perl  
libdbd-mariadb-perl  
libdbi-perl  
libfcgi-bin  
libfcgi-perl  
libfcgi0t64  
libhtml-template-perl  
libmariadb3  
libpcre2-posix3  
libterm-readkey-perl  
liburing2  
mariadb-client-core  
mariadb-common  
mariadb-plugin-provider-bzip2  
mariadb-plugin-provider-lz4  
mariadb-plugin-provider-lzma  
mariadb-plugin-provider-lzo  
mariadb-plugin-provider-snappy  
mariadb-server-core  
mysql-common  
pv  
rsync  
socat  
  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 26, Not Upgrading: 0  
  Freed space: 75.2 MB  
  
Continue? [Y/n] Y  
(Reading database ... 300078 files and directories currently installed.)
```





Base de Datos

Ajustes de instalación



- Limpiar archivos de configuración
 - `sudo rm -rf /etc/mariadb/`
- Verificar la eliminación de MariaDb
 - `mariadb`

```
developer@10: ~  
developer@10:~$ sudo rm -rf /etc/mariadb/  
developer@10:~$ mariadb  
bash: /usr/bin/mariadb: No such file or directory  
developer@10:~$
```





Base de Datos



Instalar Mariadb

- **sudo apt update**

```
developer@10: ~  
developer@10:~$ sudo apt update  
Get:1 http://security.debian.org/debian-security trixie-security InRelease [43.4  
kB]  
Hit:2 http://deb.debian.org/debian trixie InRelease  
Hit:3 http://deb.debian.org/debian trixie-updates InRelease  
Hit:4 https://packages.microsoft.com/repos/code stable InRelease  
Fetched 43.4 kB in 0s (92.1 kB/s)  
All packages are up to date.  
developer@10:~$
```





Base de Datos



Instalar Mariadb

- **sudo apt install mariadb-server**

```
developer@10: ~  
developer@10:~$ sudo apt install mariadb-server  
Installing:  
  mariadb-server  
  
Installing dependencies:  
  galera-4 mariadb-client  
  libcgi-fast-perl mariadb-client-core  
  libcgi-pm-perl mariadb-common  
  libconfig-inifiles-perl mariadb-plugin-provider-bzip2  
  libdbd-mariadb-perl mariadb-plugin-provider-lz4  
  libdbi-perl mariadb-plugin-provider-lzma  
  libfcgi-bin mariadb-plugin-provider-lzo  
  libfcgi-perl mariadb-plugin-provider-snappy  
  libfcgi0t64 mariadb-server-core  
  libhtml-template-perl mysql-common  
  libmariadb3 pv  
  libpcre2-posix3 rsync  
  libterm-readkey-perl socat  
  liburing2  
  
Suggested packages:  
  libmldbm-perl mailx openssh-server  
  libnet-daemon-perl mariadb-test python3-braceexpand  
  libsql-statement-perl netcat-openbsd  
  libipc-sharedcache-perl doc-base
```





Base de Datos



Instalar Mariadb

- **sudo apt remove mysql-apt-config**

```
developer@10: ~  
developer@10:~$ sudo mysql_secure_installation  
sudo: mysql_secure_installation: command not found  
developer@10:~$ sudo apt remove mysql-apt-config  
Package 'mysql-apt-config' is not installed, so not removed  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
developer@10:~$
```





Base de Datos



Instalar Mariadb

- **sudo find /etc/apt/sources.list.d/ -name “*mysql*” -delete**
- **sudo apt update**





Base de Datos



Validar Mariadb

Iniciar servicio

- **systemctl start mariadb**
- **systemctl enable mariadb**

Validar status

- **systemctl status mariadb**





Base de Datos



Validar Mariadb

Validar status

- **systemctl status mariadb**

```
developer@10: ~  
mariadb.service - MariaDB 11.8.3 database server  
  Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)  
  Active: active (running) since Sun 2025-11-02 05:35:21 -05; 9min ago  
 Invocation: 52883a6b5db84274ad71f1e1aaf485b1  
    Docs: man:mariabdd(8)  
          https://mariadb.com/kb/en/library/systemd/  
 Main PID: 31261 (mariabdd)  
   Status: "Taking your SQL requests now..."  
    Tasks: 9 (limit: 29735)  
  Memory: 123.8M (peak: 128.5M)  
     CPU: 3.261s  
   CGroup: /system.slice/mariadb.service  
           └─31261 /usr/sbin/mariabdd  
  
Nov 02 05:35:20 10 mariabdd[31261]: 2025-11-02 5:35:20 0 [Note] Plugin 'wsrep-'  
Nov 02 05:35:20 10 mariabdd[31261]: 2025-11-02 5:35:20 0 [Note] InnoDB: Loading  
Nov 02 05:35:20 10 mariabdd[31261]: 2025-11-02 5:35:20 0 [Note] InnoDB: Buffer  
Nov 02 05:35:20 10 mariabdd[31261]: 2025-11-02 5:35:20 0 [Note] Server socket  
Nov 02 05:35:20 10 mariabdd[31261]: 2025-11-02 5:35:20 0 [Note] mariabdd: Even  
Nov 02 05:35:20 10 mariabdd[31261]: 2025-11-02 5:35:20 0 [Note] /usr/sbin/mari  
Nov 02 05:35:20 10 mariabdd[31261]: Version: '11.8.3-MariaDB-0+deb13u1 from Deb  
Nov 02 05:35:21 10 systemd[1]: Started mariadb.service - MariaDB 11.8.3 databas  
Nov 02 05:35:21 10 /etc/mysql/debian-start[31280]: Upgrading MariaDB tables if  
Nov 02 05:35:21 10 /etc/mysql/debian-start[31294]: Checking for insecure root a
```



Base de Datos



Inicializar el servicio de Base de Datos

- **sudo mysql -u root**

```
developer@10: ~  
developer@10:~$ sudo mysql_secure_installation  
sudo: mysql_secure_installation: command not found  
developer@10:~$ sudo mysql-secure-installation  
sudo: mysql-secure-installation: command not found  
developer@10:~$ sudo mysql -u root  
Welcome to the MariaDB monitor.  Commands end with ; or \g.  
Your MariaDB connection id is 31  
Server version: 11.8.3-MariaDB-0+deb13u1 from Debian -- Please help get to 10k s  
tars at https://github.com/MariaDB/Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> █
```





Base de Datos



Inicializar el servicio de Base de Datos

- Configuración de seguridad

```
# Login to MariaDB as root (no password initially)
sudo mysql -u root

# Run these commands in the MySQL prompt:
ALTER USER 'root'@'localhost' IDENTIFIED BY 'your_secure_password_here';
DELETE FROM mysql.user WHERE User='';
DELETE FROM mysql.user WHERE User='root' AND Host NOT IN ('localhost', '127.0.0.1', '::1');
DROP DATABASE IF EXISTS test;
DELETE FROM mysql.db WHERE Db='test' OR Db='test\\_%';
FLUSH PRIVILEGES;
EXIT;
```





Base de Datos



Inicializar el servicio de Base de Datos

- Configuración de base de datos

```
developer@10:~$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 33
Server version: 11.8.3-MariaDB-0+deb13u1 from Debian -- Please help get to 10k s
tars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> -- Create database
MariaDB [(none)]> CREATE DATABASE webapp_db;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> -- Create user
MariaDB [(none)]> CREATE USER 'node_user'@'localhost' IDENTIFIED BY 'dbPass123';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> -- Grant privileges
MariaDB [(none)]> GRANT ALL PRIVILEGES ON webapp_db.* TO 'node_user'@'localhost';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> -- Apply changes
MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]>
MariaDB [(none)]> -- Verify user creation
MariaDB [(none)]> SELECT user, host FROM mysql.user;
+-----+-----+
| User           | Host           |
+-----+-----+
| mariadb.sys    | localhost      |
| mysql          | localhost      |
| node_user      | localhost      |
| root           | localhost      |
+-----+-----+
4 rows in set (0.008 sec)

MariaDB [(none)]>
MariaDB [(none)]> -- Exit
MariaDB [(none)]> EXIT;
```

CONTRASEÑA



Base de Datos



Inicializar el servicio de Base de Datos

- Configuración de base de datos

```
developer@10:~$ mariadb -u node_user -p -h localhost webapp_db
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 43
Server version: 11.8.3-MariaDB-0+deb13u1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [webapp_db]> -- Seleccionar la base de datos
MariaDB [webapp_db]> USE webapp_db;
Database changed
MariaDB [webapp_db]>
MariaDB [webapp_db]> -- Crear la tabla user
MariaDB [webapp_db]> CREATE TABLE user (
  ->   id INT NOT NULL AUTO_INCREMENT PRIMARY KEY,
  ->   name VARCHAR(100) NOT NULL,
  ->   password VARCHAR(255) NOT NULL,
  ->   status ENUM('active', 'inactive', 'suspended') DEFAULT 'active',
  ->   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  ->   updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.050 sec)

MariaDB [webapp_db]>
MariaDB [webapp_db]> -- Verificar que la tabla fue creada
MariaDB [webapp_db]> SHOW TABLES;
+-----+
| Tables_in_webapp_db |
+-----+
| user                 |
+-----+
1 row in set (0.010 sec)
```





Base de Datos



Inicializar el servicio de Base de Datos

- Insertar y Consultar Datos

```
developer@10:~$ mariadb -u node_user -p -h localhost webapp_db
Enter password:
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 44
Server version: 11.8.3-MariaDB-0+deb13u1 from Debian -- Please help get to 10k stars at https://github.com/MariaDB/Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [webapp_db]> -- Insertar usuarios de ejemplo
MariaDB [webapp_db]> INSERT INTO user (name, password, status) VALUES
-> ('Juan Pérez', '$2y$10$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'active'),
-> ('María García', '$2y$10$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'active'),
-> ('Carlos López', '$2y$10$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'inactive');
Query OK, 3 rows affected (0.006 sec)
Records: 3  Duplicates: 0  Warnings: 0

MariaDB [webapp_db]>
MariaDB [webapp_db]> -- Ver los datos insertados
MariaDB [webapp_db]> SELECT * FROM user;
```

id	name	password	status	created_at	updated_at
1	Juan Pérez	\$2y\$10\$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi	active	2025-11-02 06:22:49	2025-11-02 06:22:49
2	María García	\$2y\$10\$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi	active	2025-11-02 06:22:49	2025-11-02 06:22:49
3	Carlos López	\$2y\$10\$92IXUNpkj00r0Q5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi	inactive	2025-11-02 06:22:49	2025-11-02 06:22:49

```
3 rows in set (0.001 sec)

MariaDB [webapp_db]> █
```





Base de Datos



Conectar con Back-end

- Instalar la dependencia **mariadb** en el proyecto
nodejs
npm i mariadb



```
1 package.json > {} devDependencies
2 {
3   "name": "backend",
4   "version": "1.0.0",
5   "description": "Backend con Node.js, Express y MySQL",
6   "type": "module",
7   "main": "app.js",
8   "scripts": {
9     "start": "node app.js",
10    "dev": "nodemon app.js",
11    "test": "echo \\\"Error: no test specified\\\" && exit 1"
12  },
13  "keywords": [],
14  "author": "",
15  "license": "ISC",
16  "dependencies": {
17    "bcryptjs": "^3.0.2",
18    "cors": "^2.8.5",
19    "dotenv": "^17.2.3",
20    "express": "^5.1.0",
21    "mariadb": "^3.4.5",
22    "mysql2": "^3.15.3"
23  },
24  "devDependencies": {
25    "nodemon": "^3.1.10"
26  }
27 }
```



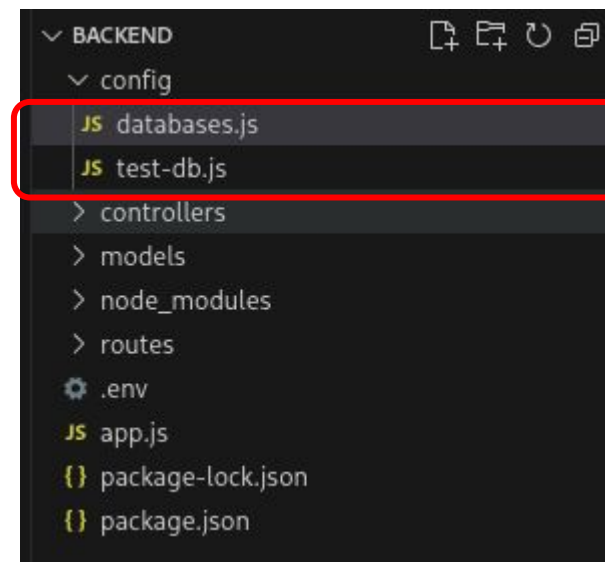


Base de Datos

Conectar con Back-end database.js



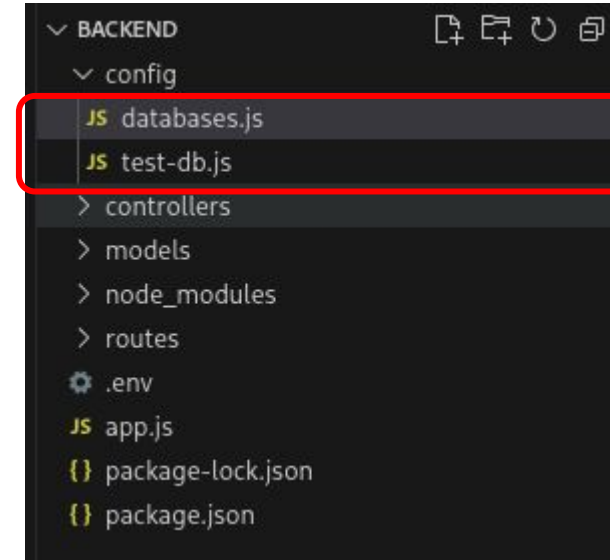
```
config > JS databases.js > testConnection
1  import mariadb from 'mariadb';
2
3  export const pool = mariadb.createPool({
4    host: process.env.DB_HOST || 'localhost',
5    user: process.env.DB_USER || 'node_user',
6    password: process.env.DB_PASSWORD || 'dbPass123',
7    database: process.env.DB_NAME || 'webapp_db',
8    port: parseInt(process.env.DB_PORT) || 3306,
9    connectionLimit: 5,
10   acquireTimeout: 30000,
11   reconnect: true,
12   timeout: 30000
13 });
14
```



Base de Datos

Conectar con Back-end database.js

```
15 export async function testConnection() {
16   let conn;
17   try {
18     conn = await pool.getConnection();
19     console.log('✅ Connected to database successfully');
20
21     // Test query to get database version
22     const result = await conn.query('SELECT VERSION() as version');
23     console.log('📄 Database Version: ${result[0].version}');
24
25     return true;
26   } catch (error) {
27     console.error('❌ Database connection failed:', error.message);
28
29     // Provide helpful debugging info
30     switch (error.code) {
31       case 'ER_ACCESS_DENIED_ERROR':
32         console.log('💡 Check username and password in database configuration');
33         break;
34       case 'ER_BAD_DB_ERROR':
35         console.log('💡 Check if database "node_app" exists');
36         break;
37       case 'ECONNREFUSED':
38         console.log('💡 Check if database server is running on localhost:3306');
39         console.log('    Run: sudo systemctl status mariadb (or mysql)');
40         break;
41       default:
42         console.log('💡 Check database server status and credentials');
43     }
44   }
45   return false;
46 } finally {
47   if (conn) conn.release();
48 }
49 }
```



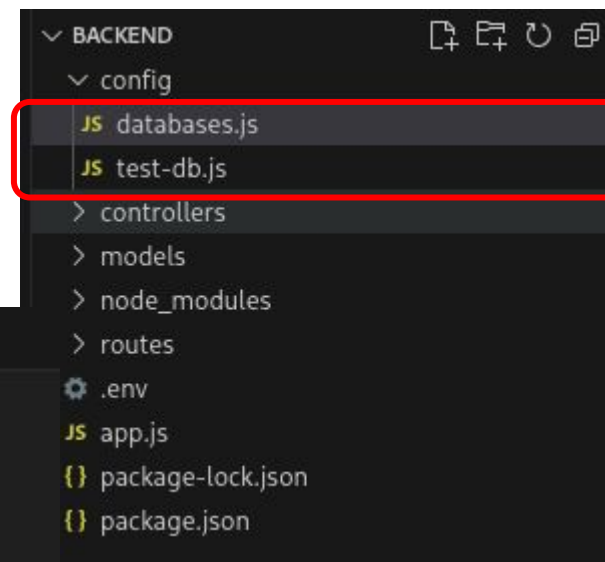


Base de Datos

Conectar con Back-end test-db.js



```
JS databases.js JS test-db.js X
config > JS test-db.js > ...
1  import { testConnection } from './databases.js';
2
3  async function test() {
4    console.log('Testing database connection...');
5    const success = await testConnection();
6    if (success) {
7      console.log('✅ All good! Database is ready for use.');
```



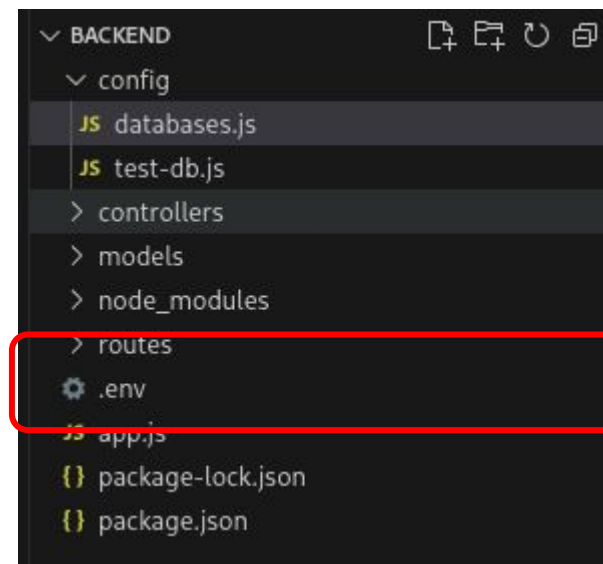


Base de Datos

Conectar con Back-end

.env

```
.env
1 DB_HOST='localhost'
2 DB_USER='node_user'
3 DB_PASSWORD='dbPass123'
4 DB_NAME='webapp_db'
5 DB_PORT='3306'
6
7 PORT='3000'
8 JWT_SECRET='KEY?SECRET'
9
10
11 MONGODB_URI=mongodb://localhost:27017/webapp_db
12 MONGODB_DB_NAME=webapp_db
13 DB_TYPE=mysql
```





Base de Datos



Conectar con Back-end

Ejecutar la validación de conexión

```
○ developer@10:~/Documents/projects/backend$ node config/test-db.js
Testing database connection...
✓ Connected to database successfully
📊 Database Version: 11.8.3-MariaDB-0+deb13u1 from Debian
✓ All good! Database is ready for use.
```





GRACIAS

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